

Block-Level Access Lists (BALs)

R&D Session at Forschungsingenieurtagung – June 2025

Duration: 1 hr

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Note taker: TBA

Agenda

- **[10 mins]** Overview and motivation for block-level access lists
- **[30 mins]** (Breakout) discussions: parallelization, zk use cases, builder complexity
- **[20 mins]** Recap and agree on path to potential fork inclusion

Summary

Block-Level Access Lists (SIP-7928) introduce explicit per-transaction access lists and state diffs into blocks. This enables deterministic parallelization of EVM execution and disk I/O, especially improving worst-case block processing time.

In addition to performance benefits, BALs support executionless validation for zkEVM nodes, improve sync/indexing, and enable new primitives like "fair" block-level warming and pre-execution validity checks. The burden of generating BALs falls on block builders, while clients and validators benefit from reduced overhead.

This session will explore the technical readiness, trade-offs, and implementation paths for BALs with the goal of assessing their suitability for inclusion in the Glamsterdam fork.

Goal(s)

- Determine consensus on whether BALs unlock enough performance and protocol value to justify inclusion in a hard fork
- Evaluate open design questions (e.g., reads vs. writes in BALs)
- Identify potential implementation blockers or coordination challenges across EL clients

Pre-reads

- SIP-7928: Block-Level Access Lists (<https://sips.sila.org/SIPS/sip-7928>)
- Block-Level Access Lists Sila Research thread

Notes

To be added after session.